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NPTEL Course on
"Computer System Optimization for ML"

Lecture #47

IISc

Agenda

- GNN Accelerator with IMC (contd.)
- Communication-aware Pruning
- IMC Accelerator for Transformer Algorithm

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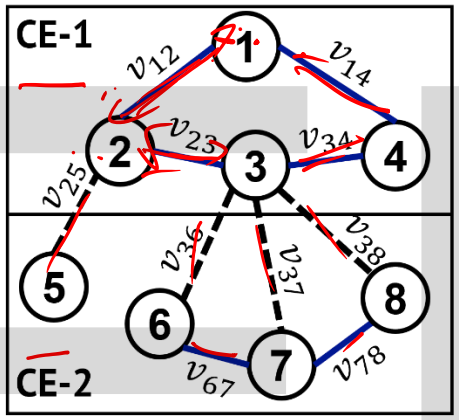
Obtaining Number of Compute Elements (CE)

- **Need to balance between inter-CE and intra-CE communication**
  - Higher number of CEs increases communication energy
  - Less number of CEs increase communication volume
- **Assume, each node generates v bits**
  - Intra-CE (solid lines): 8v in CE-1 and 4v in CE-2
  - Inter-CE (dashed lines): 8v
- **Assuming a graph with highest degree, we express the total communication energy as**

$$\sum_{i=1}^{L-1} \underbrace{\frac{N}{k}\left(\frac{N}{k}-1\right)a(i)k}_{\text{Intra-CE}} + \underbrace{\frac{N}{k}\frac{N}{k}a(i)k}_{\text{Inter-CE}}$$

: **GCN with N nodes and hardware with k CEs**

- **The expression is convex, and solved through interior point algorithm**



4 Intra-CE links $\sum_{i,j \in N_{CE-1}} v_{ij} = 8v$

4 inter-CE links $\sum_{\substack{i \in N_{CE-1} \\ j \in N_{CE-2}}} v_{ij} = 8v$

2 Intra-CE links $\sum_{i,j \in N_{CE-2}} v_{ij} = 4v$

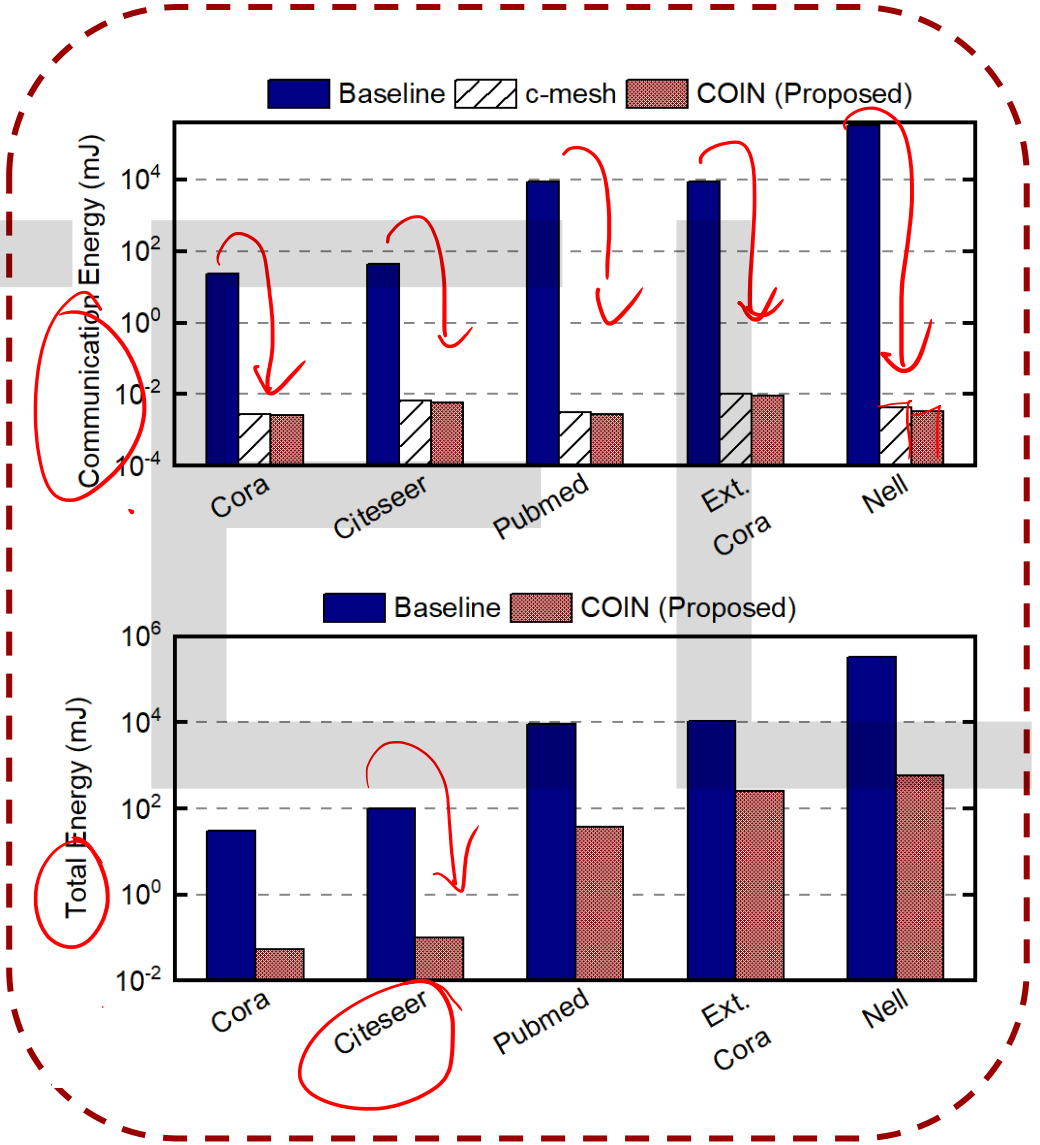
Experimental Setup

- **Performance of the computation fabric and communication fabric is estimated separately**
  - Computation fabric: NeuroSim
  - Communication fabric: BookSim
- **A wide range of GCNs is chosen**
  - Number of nodes vary between 2K-65K
  - Number of edges vary between 11K-266K
- **Baseline designs**
  - Each node is connected to a dedicated router
  - C-mesh interconnect

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Results: Improvement in Energy

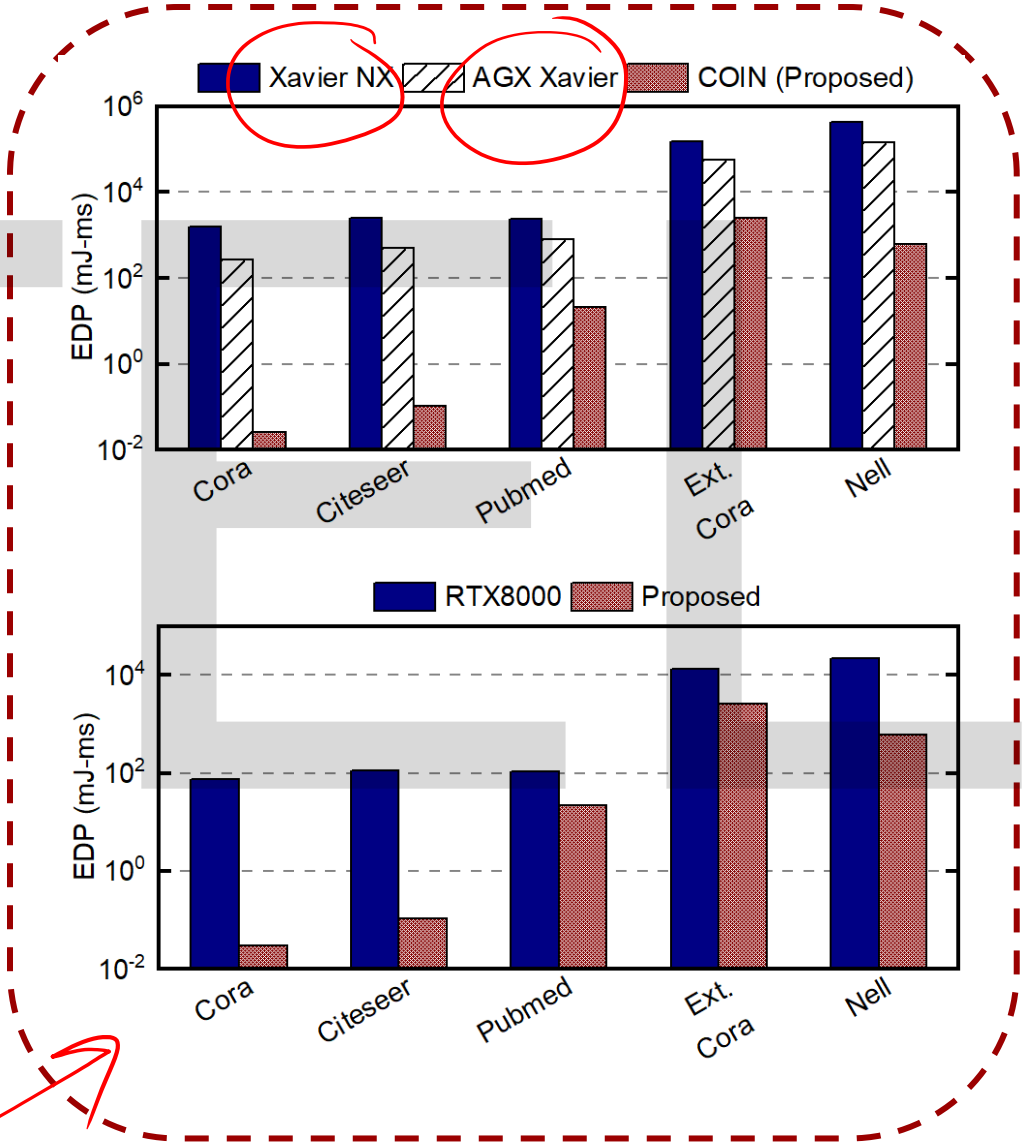
- **Up to 6 orders of magnitude improvement in communication energy w.r.t. baseline design (figure in top right)**
  - Up to 1.3× improvement w.r.t. c-mesh
- **Significant improvement in total energy consumption (figure in bottom right)**
  - Citeseer shows highest improvement (1100×)
- **Up to 350× improvement in latency**
- **Area of the chip is 17.43 mm²**



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Comparison with Existing GCN Accelerators

- **EDP Comparison with edge-devices (top)**
  - NVIDIA Jetson Xavier NX: 4 order improvement
  - NVIDIA Jetson AGX Xavier: 3 order improvement
- **EDP comparison with GPU (bottom)**
  - NVIDIA Quadro RTX 8000: 2526× improvement
- **105× improvement in energy compared to AWB-GCN [2]**



[2] Geng, Tong, et al. "AWB-GCN: A graph convolutional network accelerator with runtime workload rebalancing." *2020 53rd Annual IEEE/ACM International Symposium on Microarchitecture (MICRO)*. IEEE, 2020.

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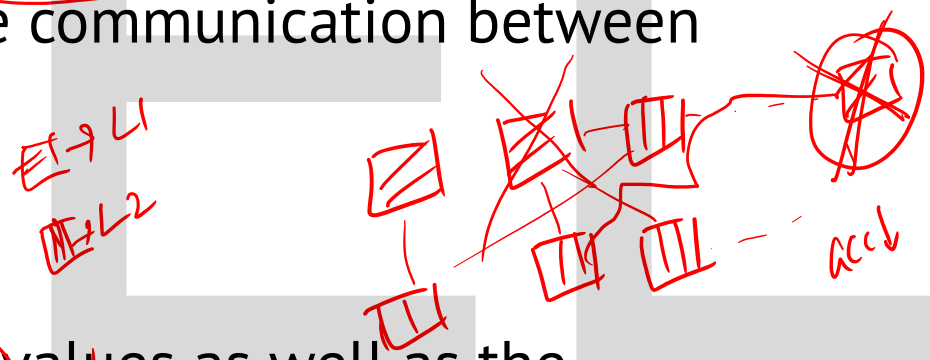
Agenda

- GNN Accelerator with IMC (contd.)
- **Communication-aware Pruning** on IMC accelerator
- IMC Accelerator for Transformer Algorithm

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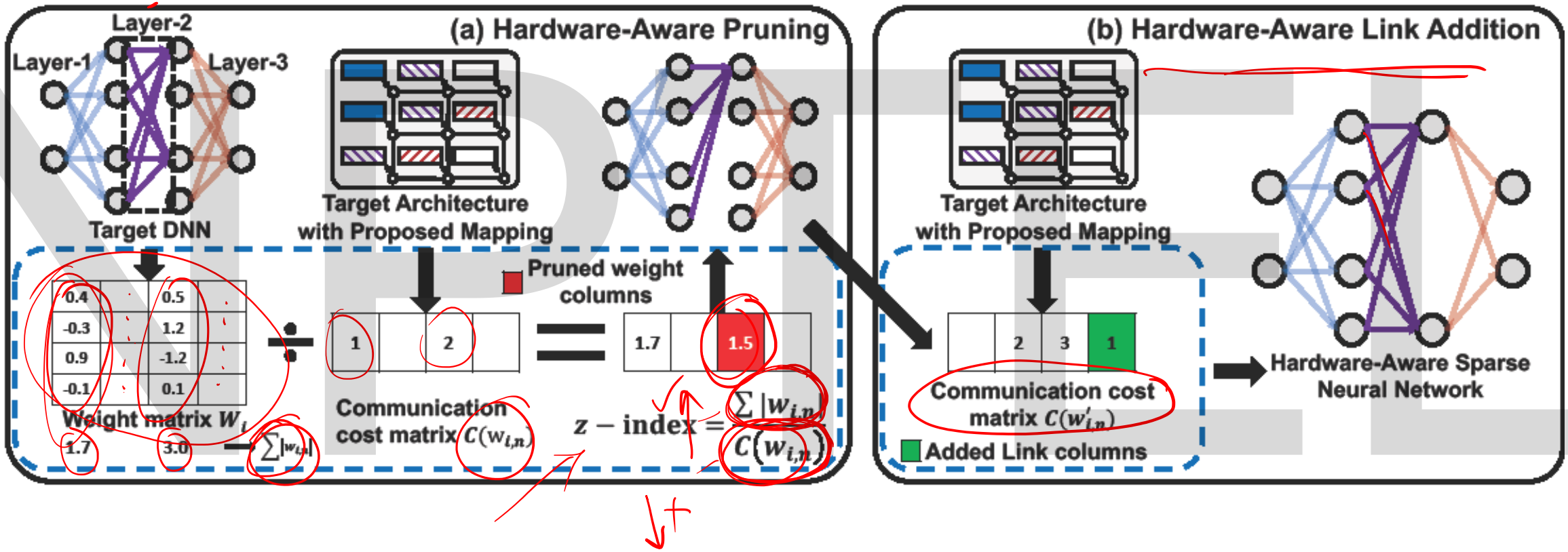
Communication-aware Pruning Technique

- **Set-up**
  - In-memory computing
  - Tiles with memory and compute
  - DNN layers (weights) are mapped to single/multiple tiles
  - Tile to tile communication takes place due to the communication between consecutive layers
- **Communication-aware pruning**
  - Pruning a filter affects the communication
  - Idea: prune those filters which has lower weight values as well as the corresponding activations travel long distance on chip
  - Iterative approach



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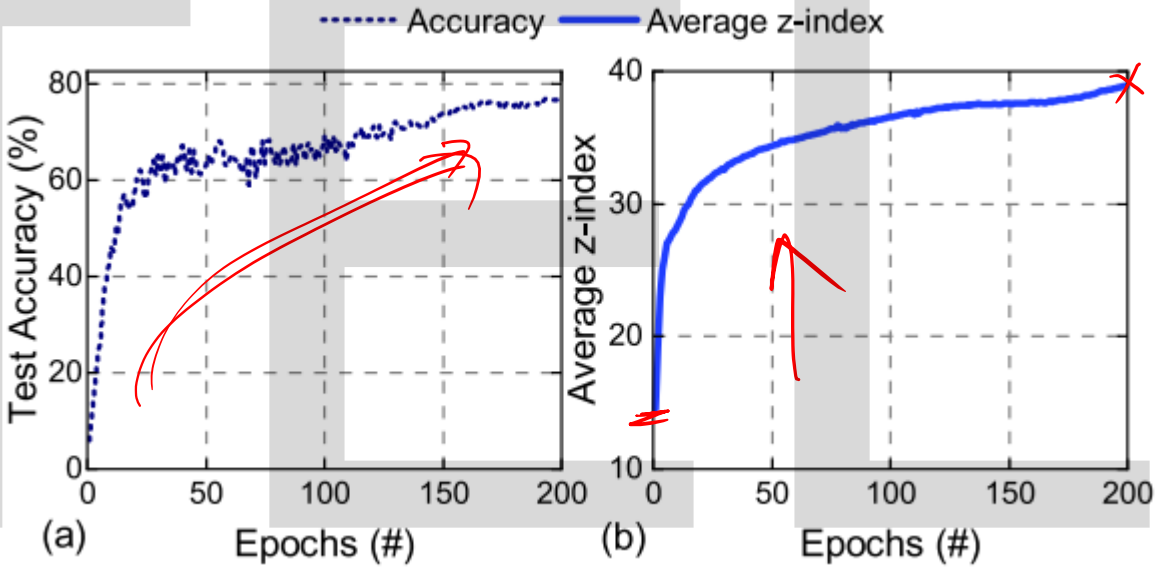
Communication-aware Pruning Technique



Ref: CANNON: Communication-Aware Sparse Neural Network Optimization [Goksoy et al]

Progression of z-index while Pruning

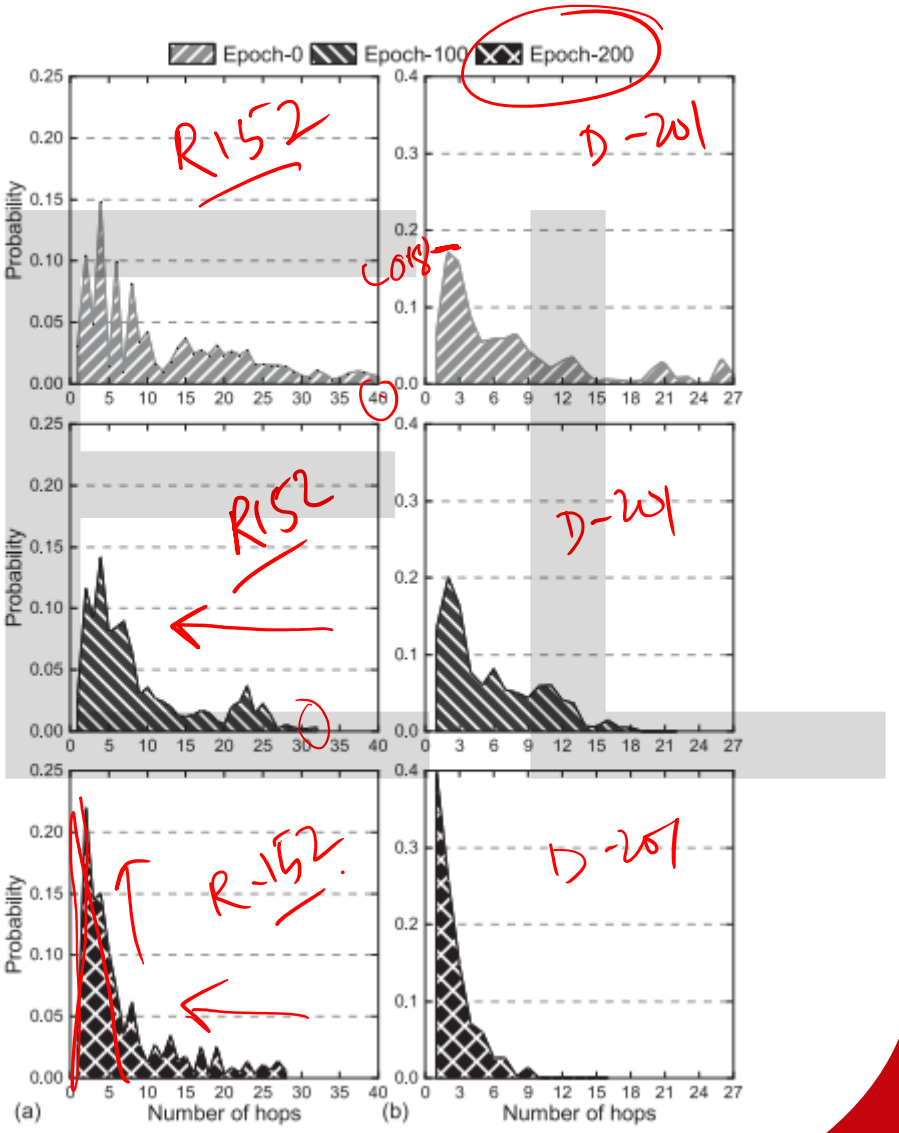
- Z-index = $\frac{weight\ value}{communication\ cost}$
- More is better
- Result is shown for ResNet 152
- As the epochs are progressing, z-index is also increasing



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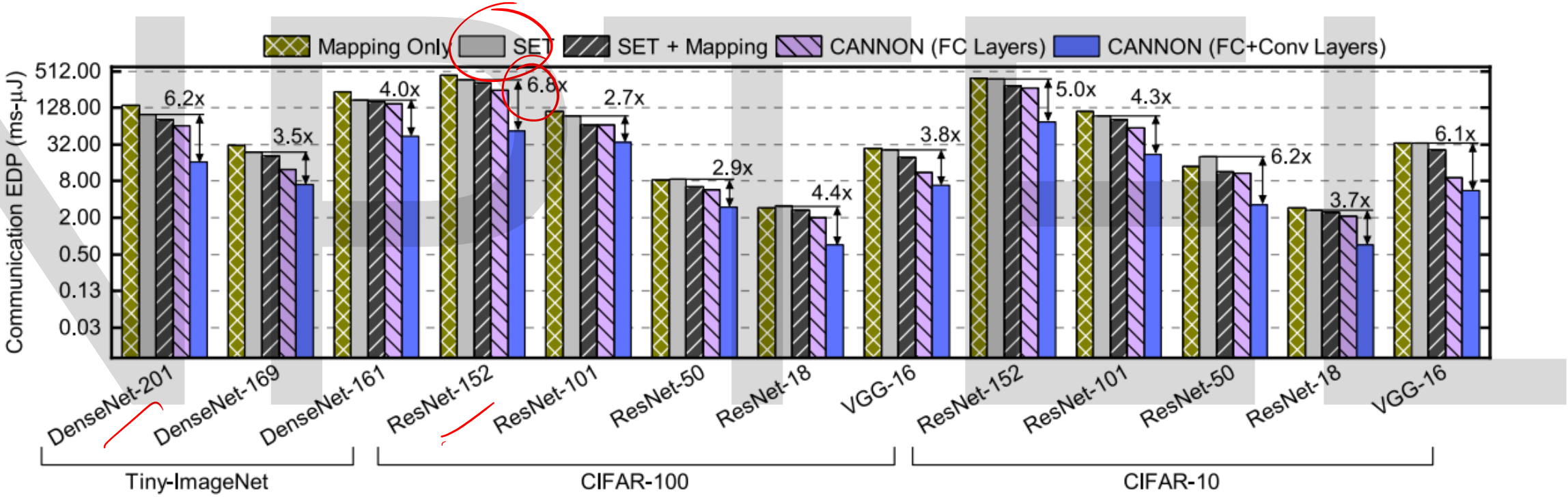
Progression of #hop Distribution

- (a) is for ResNet-152
- (b) is for DenseNet-201



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Results for Different DNNs with Different Techniques



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- Communication-aware Pruning
- **IMC Accelerator for Transformer Algorithm**

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THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)